

Abstract algebra – 7

Generators and relations: infinite and matrix groups

Homework

24 сентября 2026 года

Theoretical minimum. A presentation $\langle S \mid R \rangle$ is the free group $F(S)$ of reduced words modulo the normal subgroup generated by R ; by von Dyck's theorem every group generated by elements satisfying R is a quotient of it. Thus $D_n = \langle s, t \mid s^2 = t^2 = (st)^n = e \rangle$, $\mathbb{Z}_2 * \mathbb{Z}_2 = \langle s, t \mid s^2 = t^2 = e \rangle$ is the infinite dihedral group, and S_n is the Coxeter group with $s_i^2 = (s_i s_{i+1})^3 = (s_i s_j)^2 = e$, $|i - j| \geq 2$. Quotients and finite-index subgroups of finitely generated groups are finitely generated; arbitrary subgroups need not be. Ping-pong: if $a^k X_2 \subseteq X_1$ and $b^k X_1 \subseteq X_2$ for $k \neq 0$ and $X_1 \cap X_2 = \emptyset$, then $\langle a, b \rangle$ is free. An element is a product of two involutions if and only if an involution conjugates it to its inverse; two reflections compose to a rotation by twice the angle between the mirrors. Minkowski: an integer matrix of finite order that is $\equiv I$ modulo an odd prime p is I , so finite subgroups of $\mathrm{GL}_n(\mathbb{Z})$ embed into $\mathrm{GL}_n(\mathbb{F}_p)$. Burnside: subgroups of $\mathrm{GL}_n(\mathbb{C})$ of finite exponent are finite.

1. Prove that the composition of the reflections in two lines meeting at a point O at angle θ is the rotation about O by 2θ . Deduce that every element of the orthogonal group $O(n)$ is a product of two involutions lying in $O(n)$, and that the reflections in the sides of an equilateral triangle generate an infinite group. Show that in the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ the element i is not a product of two involutions.
2. Let the multiplicative group F^* of a field F be finitely generated. Must F^* be cyclic? (МФТИ 2024)
3. Let G be a group such that every element of its commutator subgroup G' (the subgroup generated by all $xyx^{-1}y^{-1}$) has finite order. Prove

that the set of all elements of G of finite order is a subgroup of G .

(IMS 2006)

4. Let a, b, c be elements of finite order in some group. Prove that if $a^{-1}ba = b^2$, $b^{-2}cb^2 = c^2$ and $c^{-3}ac^3 = a^2$, then $a = b = c = e$, where e is the unit element. (VJIMC 2011)
5. Let p be an odd prime. (a) Prove Minkowski's theorem and deduce that the order of a finite subgroup of $\mathrm{GL}_n(\mathbb{Z})$ divides both $|\mathrm{GL}_n(\mathbb{F}_3)|$ and $|\mathrm{GL}_n(\mathbb{F}_5)|$. (b) Prove that every finite subgroup G of $\mathrm{GL}_n(\mathbb{Q})$ is conjugate to a subgroup of $\mathrm{GL}_n(\mathbb{Z})$ (consider the subgroup $\sum_{g \in G} g\mathbb{Z}^n$ of $\mathbb{Q}^n$). (c) Find finite subgroups of $\mathrm{GL}_2(\mathbb{Z})$ of orders 8 and 12, and compare with the bound of (a).
6. Let $G \subseteq \mathrm{GL}_n(\mathbb{C})$ be a group with $g^m = I$ for all $g \in G$. Prove that $|G| \leq m^{n^3}$, assuming Burnside's theorem that an irreducible subgroup of $\mathrm{GL}_n(\mathbb{C})$ spans $M_n(\mathbb{C})$. Show that G can be infinite if (a) the exponents are allowed to depend on g ; (b) $\mathbb{C}$ is replaced by a field of characteristic p .
7. Let F_2 be the free group with two generators. Prove that the commutator subgroup of F_2 is not finitely generated. (Высшая лига 2016)
8. Prove that the group $\mathrm{SO}(3)$ of proper rotations of three-dimensional space contains a free subgroup generated by two elements. *Hint:* try the rotations by $\arccos \frac{3}{5}$ about two perpendicular axes, multiply a word of length k in them by 5^k and reduce modulo 5. (МФТИ)
9. Let p be a prime, and consider the group G consisting of 2×2 matrices $\begin{pmatrix} y^m & y^n f(y) \\ 0 & 1 \end{pmatrix}$, where f varies in $\mathbb{F}_p[y]$ and m, n over integers. (i) Show that G is finitely generated; (ii) show that there does not exist a finite sequence of elements $g_1, \dots, g_n$ (not necessarily distinct) of G such that $G = \{g_1^{k_1} g_2^{k_2} \dots g_n^{k_n} : k_i \in \mathbb{Z}\}$. (IMC 2025, shortlist)
10. Let G be the free product $\mathbb{Z}_3 * \mathbb{Z}_3 = \langle x, y \mid x^3 = y^3 = e \rangle$. If N_n denotes the number of subgroups of index n in G , prove that N_n is odd if, and only if, n is of the form $2^k - 3$. (Compare Abstract algebra – 4.) (IMC 2025, shortlist)